A Reflexive Banach Space Which Does Not Coarsely Embed Into Any Stable Metric Space

Run `fa_banach_001`, lane 9

June 14, 2026

Source Problem

The source is Braga–Swift, *Coarse embeddings into superstable spaces*, arXiv:1704.04468 [1]. On page 3 they recall Kalton's Problem 6.1 and state:

COARSE EMBEDDINGS INTO SUPERSTABLE SPACES 3

embedding. In the same paper, Kalton asked if the converse of this result also holds. Precisely, the following is open (see [K, Problem 6.1]).

Problem 1.5. Does every (separable) reflexive Banach space embed coarsely (resp. uniformly) into a stable space?

By Raynaud's result, it is clear that there are separable reflexive spaces which do not uniformly embed into superstable spaces. However, to the best of our knowledge, it was unknown whether every reflexive Banach space coarsely embeds

Thus the question is whether every separable reflexive Banach space embeds coarsely, respectively uniformly, into a stable metric space.

Result

Theorem 1. *Tsirelson's original space T^* is a separable reflexive Banach space which does not coarsely embed into any stable metric space. Consequently it gives a negative answer to Problem 1.5 of [1]. Moreover, T^* does not uniformly embed into any stable metric space.*

Proof Intuition

The obstruction is not linear containment of c_0, since T^* is reflexive and contains no copy of c_0. The obstruction is finite representability. If X finitely represents c_0, then every finite part of the integer lattice of c_0 appears in X with uniformly bounded distortion. A coarse embedding $X \to M$ into a stable metric space would therefore send those finite lattice pieces into M with uniform coarse control. Passing to a cofinal metric ultrapower turns these finite pieces into a genuine coarse embedding of a net in c_0 into a stable metric space, contradicting Kalton's theorem that c_0 does not coarsely embed into any stable metric space.

Proof

We use two standard facts. First, metric stability is preserved by metric ultrapowers. Second, Kalton proved that c_0 does not coarsely or uniformly embed into a stable metric space [3]; this is also quoted in [1, p. 3].

Lemma 1. *Let X be a Banach space which finitely represents c_0. Then X does not coarsely embed into any stable metric space.*

Proof. Assume toward a contradiction that $f : X \to M$ is a coarse embedding into a stable metric space M. Let $\rho, \omega : [0, \infty) \to [0, \infty)$ be nondecreasing control functions with

$$\rho(\|x - y\|) \leq d_M(f(x), f(y)) \leq \omega(\|x - y\|) \quad (x, y \in X),$$

$\rho(t) \to \infty$, and $\omega(t) < \infty$ for every t.

Let

$$\Lambda = c_0 \cap \mathbb{Z}^{\mathbb{N}} = c_{00}(\mathbb{Z})$$

with the c_0-metric. This lattice is a net in c_0: rounding each coordinate of $z \in c_0$ to a nearest integer gives an element of Λ within distance $1/2$.

Let I be the directed set of finite subsets of Λ, ordered by inclusion, and let $\mathcal{U}$ be an ultrafilter on I extending the cofinal filter, so that $\{E \in I : F \subset E\} \in \mathcal{U}$ for every finite $F \subset \Lambda$. Since c_0 is finitely represented in X, after fixing a distortion constant $D \geq 1$, for each $E \in I$ there are $n(E) \in \mathbb{N}$, containing the supports of all elements of E, and a linear map

$$T_E : \ell_\infty^{n(E)} \longrightarrow X$$

such that

$$\|u\|_\infty \leq \|T_E u\| \leq D\|u\|_\infty \quad (u \in \ell_\infty^{n(E)}).$$

For $a \in \Lambda$, put $x_E(a) = T_E(P_{n(E)}a)$, where P_n is the first n-coordinate projection. The family $(f(x_E(a)))_{E \in I}$ is bounded in M, because $\|x_E(a)\| \leq D\|a\|_\infty$. Therefore it defines a point

$$\Phi(a) = [\, f(x_E(a))\,]_{\mathcal{U}}$$

in the metric ultrapower $M_{\mathcal{U}}$.

For fixed $a, b \in \Lambda$, the set of E's containing both a and b belongs to $\mathcal{U}$. On this set,

$$\|a - b\|_\infty \leq \|x_E(a) - x_E(b)\| \leq D\|a - b\|_\infty.$$

Taking the $\mathcal{U}$-limit in M, we get

$$\rho(\|a - b\|_\infty) \leq d_{M_{\mathcal{U}}}(\Phi(a), \Phi(b)) \leq \omega(D\|a - b\|_\infty).$$

Thus $\Phi : \Lambda \to M_{\mathcal{U}}$ is a coarse embedding. Since Λ is a net in c_0, choose a nearest-lattice map $q : c_0 \to \Lambda$. Then

$$\|x - y\|_\infty - 1 \leq \|q(x) - q(y)\|_\infty \leq \|x - y\|_\infty + 1,$$

so $\Phi \circ q$ is a coarse embedding of c_0 into $M_{\mathcal{U}}$. But $M_{\mathcal{U}}$ is stable, contradicting Kalton's non-embeddability theorem for c_0. $\square$

Lemma 2. *Let X be a Banach space which finitely represents c_0. Then X does not uniformly embed into any stable metric space.*

Proof. If $f : X \to M$ were a uniform embedding into a stable metric space, then the usual Banach ultrapower argument would give a uniform embedding of c_0 into a stable metric ultrapower of M. Indeed, finite representability of c_0 in X gives an isomorphic embedding $J : c_0 \to X_{\mathcal{U}}$ into a Banach ultrapower of X. Uniform continuity of f, together with the fact that X is a normed vector space, implies that f maps bounded subsets of X to bounded subsets of M and induces a uniformly continuous map

$$f_{\mathcal{U}} : X_{\mathcal{U}} \to M_{\mathcal{U}}.$$

The inverse uniform modulus of f passes to $f_{\mathcal{U}}$: if $\|u - v\| \geq \varepsilon$ in $X_{\mathcal{U}}$, then

$$d_{M_{\mathcal{U}}}(f_{\mathcal{U}}u, f_{\mathcal{U}}v) \geq \rho_f(\varepsilon/2) > 0,$$

where $\rho_f(t) = \inf\{d_M(f(x), f(y)) : \|x - y\| \geq t\}$. Hence $f_{\mathcal{U}} \circ J$ is a uniform embedding of c_0 into the stable metric space $M_{\mathcal{U}}$, again contradicting Kalton's theorem. □

We now apply the lemmas to Tsirelson's original space. Baudier–Lancien– Schlumprecht recall that T^* is reflexive and that all spreading models of T^* are isomorphic to c_0 [2, p. 18]. Therefore c_0 is finitely represented in T^*. Lemmas 1 and 2 show that T^* neither coarsely nor uniformly embeds into any stable metric space. Since T^* is separable and reflexive, this proves Theorem 1. □

Verification and Novelty Check

The cheap run indexes were searched for `1704.04468`, `ProblemReflexEmbStable`, `stable metric`, `separable reflexive`, `original Tsirelson`, and `c_0 spreading`; no duplicate packet or attempt for this target was found. Web searches for exact and close variants of the source problem and the obstruction, including

"Does every (separable) reflexive Banach space embed coarsely",

"stable metric space" "Tsirelson", "c_0" "stable metric" "coarse",

found no prior answer. arXiv API searches for the corresponding phrase combinations returned zero results. The later Baudier–Lancien–Schlumprecht paper is adjacent literature, but it is used here only for the structural facts about T^*; it does not explicitly answer Problem 1.5 of [1].

The main point for human review is Lemma 1, especially the cofinal-ultrapower construction over finite subsets of the integer lattice of c_0, and the standard closure of metric stability under metric ultrapowers.

References

[1] B. M. Braga and A. Swift, *Coarse embeddings into superstable spaces*, arXiv:1704.04468, 2017.

[2] F. Baudier, G. Lancien, and Th. Schlumprecht, *The coarse geometry of Tsirelson's space and applications*, arXiv:1705.06797, 2017.

[3] N. J. Kalton, *Coarse and uniform embeddings into reflexive spaces*, Quarterly Journal of Mathematics 58 (2007), 393–414.